

biochemistry1

1- As the pKa of an acid increase, the acid will be:

Stronger



Weaker

Converted to basic

Converted to neutral

2- Dipalmitoyl lecithin acts as



Lung surfactant

Platelet activating factor

Second messenger for hormones

Anti-ketogenic compound

3- The proofreading function of DNA polymerase involves all of the following except

Base pairing



Reversal of the polymerization reaction

Detection of mismatched base pairs

A 3' → 5' exonuclease

4- When double-stranded DNA is heated at neutral pH, which change does not occur?

The helical structure unwinds



The covalent N-glycosidic bond between the base and the pentose breaks

The absorption of ultraviolet (260 nm) light increases

The hydrogen bonds between A and T break

5- The operator region normally can be bound by



Repressor

Suppressor tRNA

mRNA

Inducer

6- The gene of lac operon which has constitutive expression is.....

c

p

z

✓ i

7- Which of the following fatty acids has the lowest melting point?

Palmitic acid

✓ Eicosapentaenoic acid

Arachidonic acid

Stearic acid

8- Deficiency of which one of these vitamins is the major cause of blindness?

Vitamin D

Vitamin B6

Vitamin B12

✓ Vitamin A

9- All of the following statements correctly describes transport of O₂ by hemoglobin except

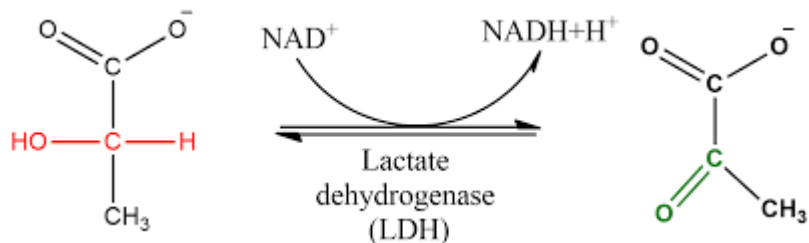
The plot of percentage of O₂ bound versus O₂ pressure is sigmoidal in shape

CO binds to hemoglobin more avidly than does O₂

All of the four heme moieties binds O₂ cooperatively

✓ Increased CO₂ concentrations increase O₂ affinity

10- Which class of enzymes may catalyze the following reaction?



- ✓ Oxidoreductases
- Ligases
- Hydrolases
- Isomerases

11- Which one of the following results in Ehlers-Danlos syndrome?

- ✓ Deficiency of lysyl hydroxylase
- Deficiency of elastase
- Deficiency of vitamin C
- Deficiency of antitrypsin

12- Which of the following is not an example of a weak acid?

- Lactic acid
- Pyruvic acid
- ✓ Sulfuric acid
- Carbonic acid

13- Histones are _____ that are usually associated with _____.

- ☐ Acidic proteins -RNA
- ☒ Basic proteins - DNA
- ☐ Acidic proteins -DNA
- ☐ Basic proteins -RNA

14- Ciprofloxacin inhibits.....

- ☐ DNA polymerase III
- ☒ DNA gyrase
- ☐ DNA polymerase I
- ☐ DNA topoisomerase I

15- In bacteria the elongation stage of protein synthesis does not involve

- ☒ IF-2
- ☐ Peptidyl transferase
- ☐ GTP
- ☐ EF-Tu

16- The catalytic efficiency of two different enzymes can be compared by their

- ☐ Product type
- ☐ Optimum pH value
- ☐ Molecular size of the enzymes
- ☒ Km values

17- Formic acid is also called

- ☐ Formaldehyde
- ☐ Ethanoic acid
- ☐ Acetone
- ☒ Methanoic acid

18- The general formula of amines is

- ✓ ☐ R-NH₂
- ☐ R-COOH
- ☐ R-CH₂
- ☐ R₂CH

19- Choose the incorrect statement out of the following.

- ☐ Glycine is optically inactive
- ☐ Selenocysteine is 21st amino acid
- ☐ Only L-amino acids are found in the biological system
- ✓ ☒ Tyrosine is a modified amino acid

20- In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is ...

- ✓ ☒ Substitution of a single amino acid
- ☐ Reduced affinity for oxygen
- ☐ Faulty binding of the heme groups
- ☐ Incorrect secondary structure

21- Which one of the following is a reversible inhibitor of the COX-1 enzyme that is involved in prostaglandin synthesis?

- ☐ Sarin
- ☐ Penicillin
- ✓ ☒ Ibuprofen
- ☐ Aspirin

22- Which of the following produces N-acetyl glucosamine upon hydrolysis?

- ☐ Dextran
- ✓ ☒ Chitin
- ☐ Cellulose
- ☐ Inulin

23- The strength of an organic acid depends on:

- ☐ Number of neutrons gained
- ☐ Number of hydroxyl released
- ☒ Electronegativity
- ☐ Number of double bonds

24- Amylopectin has:

- ☐ $\beta(1 \rightarrow 4)$ and $\alpha(1 \rightarrow 6)$ linkage
- ☐ $\beta(1 \rightarrow 2)$ linkage
- ☒ $\alpha(1 \rightarrow 4)$ and $\alpha(1 \rightarrow 6)$ linkage
- ☐ $\alpha(1 \rightarrow 2)$ linkage

25- Which of the following is a pair of non-reducing carbohydrate?

- ☒ Sucrose and trehalose
- ☐ Lactose and maltose
- ☐ Cellobiose and sucrose
- ☐ Trehalose and maltose

26- Polyprotic acids such as H_3PO_4 , can act as acid-base buffers....

- ☐ At pH values around neutrality
- ☒ At pH values around any of their pKa's
- ☐ If their concentration is kept low
- ☐ Only in combination with polyprotic bases

27- Which vitamin is a major lipid-soluble antioxidant in cell membrane?

- ☒ Vitamin E
- ☐ Vitamin K
- ☐ Vitamin C
- ☐ Vitamin D

28- All of the following vitamins provide the cofactor for pyruvate dehydrogenase, except...

Riboflavin

Niacin



Folate

Thiamine

29- A buffer solution contains propionic acid and its conjugate base; the pK_a of propionic acid is 4.88. At what pH would the solution buffer best?

3.0



5.0

4.0

6.0

30- Which of the following disaccharides contains a beta-glycosidic bond?



Lactose and cellobiose

Maltose and sucrose

Maltose and amylose

Lactose and sucrose

31- A non-competitive inhibitor of an enzyme ...

Increases both K_m and V_{max}

Increases K_m with no change in V_{max}



Decreases V_{max} with no change in K_m

Decreases both V_{max} and K_m

32- Which one of the following is assembled and secreted by the liver?

Chylomicrons

High Density Lipoprotein (HDL)



Very Low-Density Lipoprotein (VLDL)

Low Density Lipoprotein (LDL)

33- The compound that consists of ribose linked by an N-glycosidic bond to N-9 of adenine is

Adenosine monophosphate

A purine nucleotide

A deoxyribonucleoside

✓ Adenosine

34- Non-steroidal anti-inflammatory drugs (NSAIDs) like aspirin and ibuprofen act by blocking the production of

Vitamin D

✓ Prostaglandins

Biological waxes

Sphingolipids

35- Which of the following statements concerning fatty acids is correct?

They are strongly hydrophilic

They all contain one or more double bond

✓ One of them is a precursor of prostaglandins

They are a constituent of sterol

36- Oligosaccharides linked to proteins are called:

Glucosides

Glycosaminoglycans

✓ Glycoproteins

Glycolipids

37- In polysaccharides, monosaccharides are joined by:

Glucose bonds

Ester bonds

✓ Glycosidic bonds

Peptide bonds

38- An amino acid not found in proteins is ...

Histidine



β -Alanine

Proline

Lysine

39- RNA polymerase from E. coli (core enzyme alone) has all of the following properties except that it

Is required for the synthesis of mrna, rrna, and trna in E. Coli

Produces an RNA polymer that begins with a 5'-triphosphate

Can extend an RNA chain and initiate a new chain



Recognizes specific start signals in DNA

40- In base-excision repair, the first enzyme to act is

DNA ligase

Dam methylase

DNA polymerase



DNA glycosylase

41- The secondary structure of protein is held together by which type of bonding?

3-D stereochemical folding



Hydrogen bonding

Covalent bonding

Spatial arrangement

42- Tetracycline prevents synthesis of polypeptide by:



Preventing binding of aminoacyl tRNA

Blocking mRNA formation from DNA

Competing with mRNA for ribosomal binding sites

Releasing peptides from mRNA-tRNA complex

43- The enzyme which hydrolyses Glucose-Fructose linkage is....

Maltase



Sucrase

Amylase

Lactase

44- Which vitamin requires cobalt for its activity?

Vitamin A

Vitamin B2

Vitamin D



Vitamin B12

45- Which is incorrect statement about vitamin C?



It can be synthesized in the body

It shows antioxidant activity

It is a strong reducing agent

It is involved in hydroxylation of prolyl- and lysyl- residues of collagen

46- Sphingosine is not a component of....



Cardiolipin

Ceramide

Sphingomyelin

Gangliosides

47- Deficiency of which one of these vitamins may lead to megaloblastic anemia?

Vitamin E

Vitamin D



Vitamin B12

Vitamin B6

48- Which of the followings inhibit the DNA-dependent RNA polymerase in prokaryotes?

- ✓ Rifampicin
- Chloramphenicol
- cycloheximide
- Ciprofloxacin

49- Regarding protein synthesis in eukaryotes, all proteins are initially synthesized with?

- Methionine at their C-terminus
- Tryptophan at their C-terminus
- A multiple of 3 amino acids in their sequence
- ✓ Methionine at their N-terminus

50- In scurvy, which amino acid that is normally a part of collagen is not synthesized?

- ✓ Hydroxyproline
- Hydroxytryptophan
- Hydroxyalanine
- Hydroxytyrosine

51- Which of the following has more than one type of sugar in its structure?

- ✓ Hyluronic acid
- Inulin
- Amylopectin
- Glycogen

52- Which of the following contains an ether-linked alkyl group?

- ✓ Platelet-activating factor
- Gangliosides
- Phosphatidyl serine
- Sphingomyelin

53- Structural polysaccharides include:

Chitin and glycogen

Cellulose and starch



Cellulose and chitin

Glycogen and starch

54- If you replace the oxygen of the hydroxyl group with sulfur in R-OH, which functional group will you get?

hydroxyl

Sulfa



Thiol

Sulfoxyl

55- Which of the following vitamins provides the cofactor for transamination of amino acids?

Niacin

Retinol

Phylloquinone



Pyridoxine

56- Which of the following can cancel the effect of an irreversible inhibitor?

Increasing the substrate concentration



Removing the inhibitor

Increasing the product concentration

Increasing the temperature

57- Which one of the following antibiotics does not function by interfering with the translational process?



Penicillin

Cycloheximide

Streptomycin

Puromycin

58- Which of the following would produce Methemoglobinemia?

Transition of Glutamic acid to Valine

Deletion of one β -globin locus

Deletion of one α -globin locus

✓ Converting Ferrous iron (Fe^{2+}) to Ferric (Fe^{3+})

59- Glycine and proline are the most abundant amino acids in the structure of

Hemoglobin

Insulin

Myoglobin

✓ Collagen

60- The lipoprotein richest in cholesterol is ...

Chylomicron

HDL

✓ LDL

VLDL

61- When $K_m = S$, reaction velocity (v) would be equal to.....

$3/4 V_{max}$

$1 V_{max}$

$2 V_{max}$

✓ $1/2 V_{max}$

62- Thiamine deficiency leads to_____?

Pellagra

Scurvy

Rickets

✓ Beri-beri

63- The tryptophan operon of E. coli is repressed by tryptophan added to the growth medium. The tryptophan repressor probably

Binds To The Trp Operator In The Absence Of Tryptophan

Is an attenuator



Binds to the trp operator in the presence of tryptophan

Binds to RNA polymerase when tryptophan is present

64- When a DNA molecule is described as replicating bidirectionally, it means that it has two ...

Termination points



Replication forks

Origins

Independently replicating segment